Fatigue Monitoring using Wearable Technology

AUTHORS

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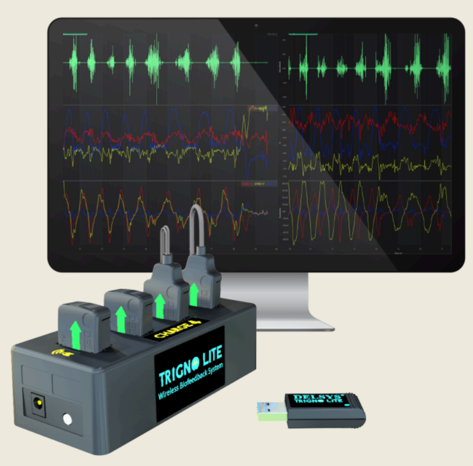
This study investigates the use of wearable EMG sensors to monitor and predict muscle fatigue in workers performing repetitive forearm tasks to enhance safety and productivity in advanced manufacturing and supply chain environments.

01. Introduction

In Advanced Manufacturing and Supply Chain (AMSC), repetitive tasks often cause significant muscle fatigue, compromising worker safety and productivity.

This study aims to explore the feasibility of using wearable electromyography (EMG) sensors to monitor and predict muscle fatigue in workers performing repetitive forearm tasks, a common source of workplace injuries.

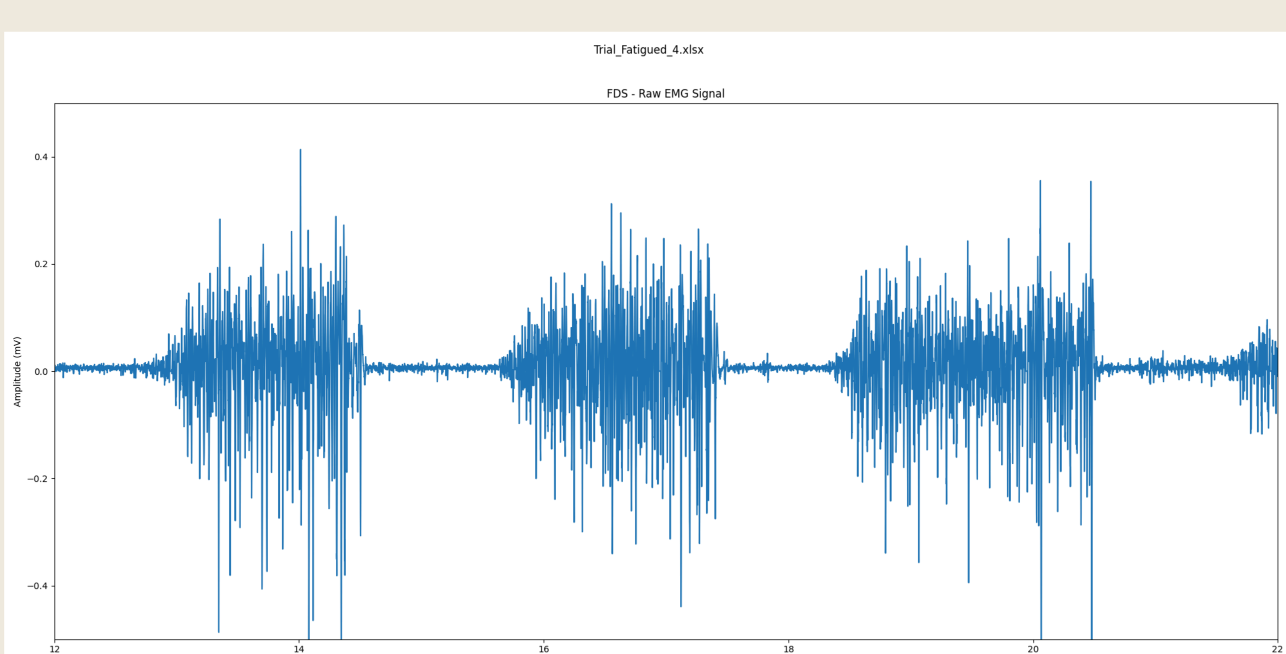
By developing effective signal processing techniques and analyzing various fatigue indicators, we aim to enhance worker safety and productivity though proactive fatigue management.



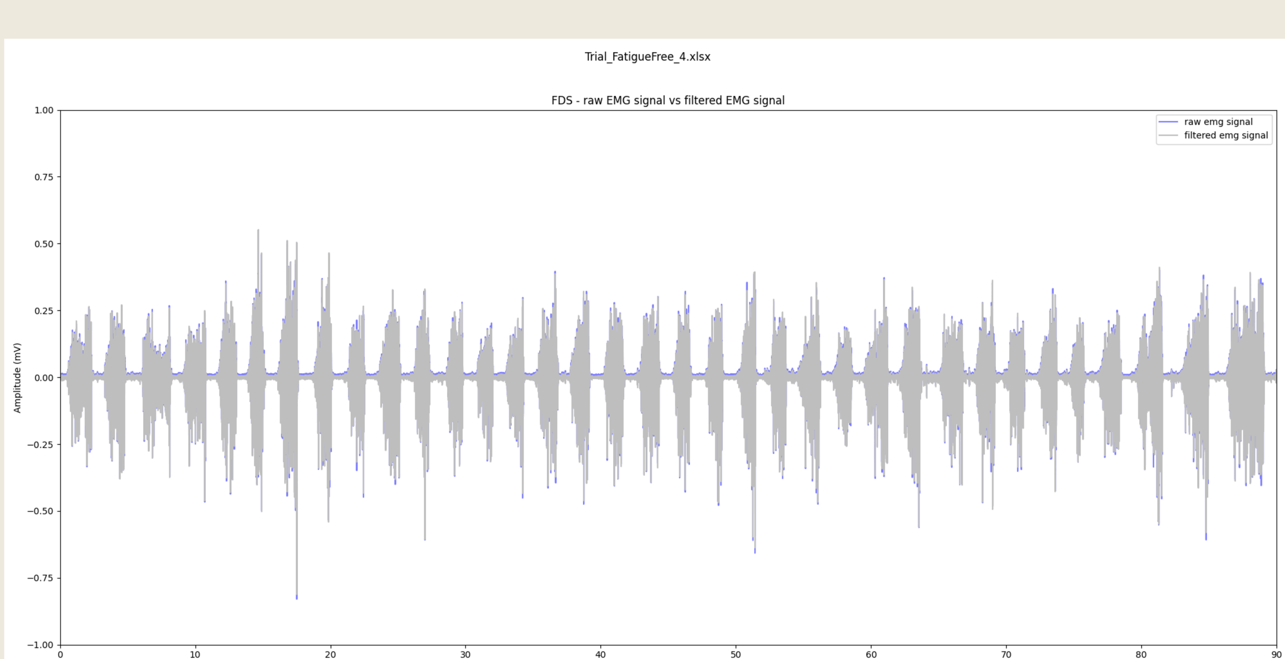
02. Methodology

The muscle activity from the forearm muscles Flexor Digitorum and Extensor Digitorum were recorded.

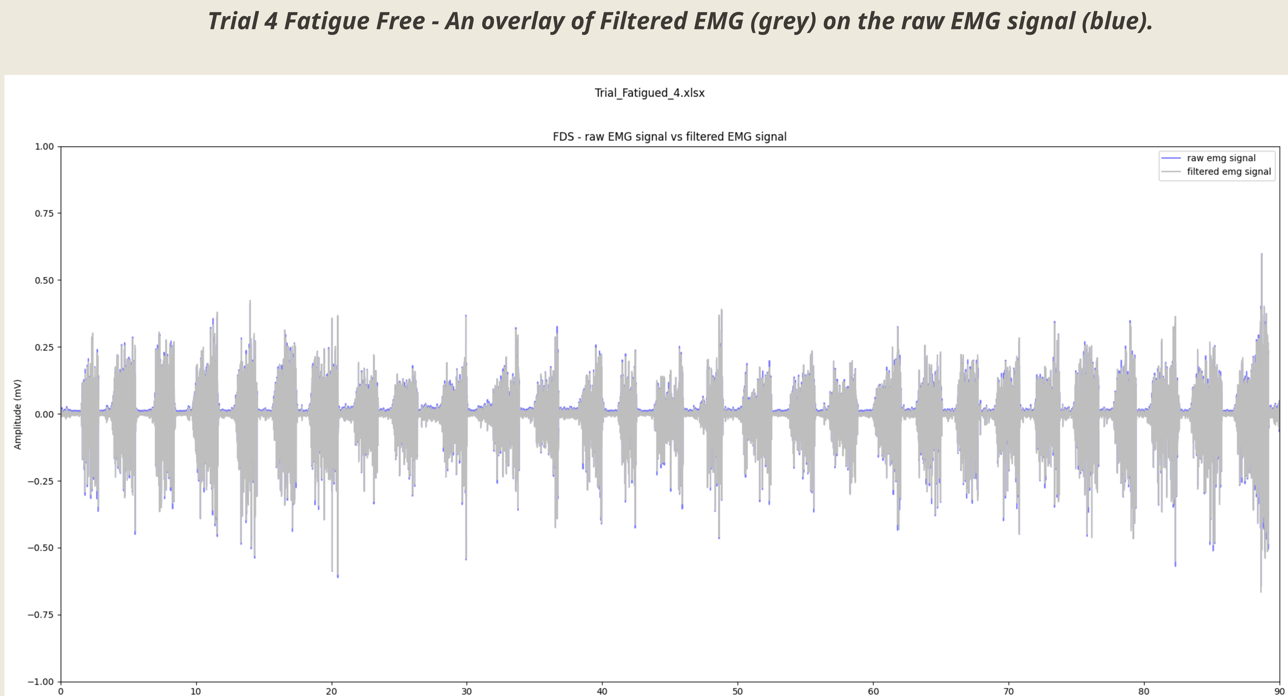
- **Baseline Data Collection:** I conducted 5 trials involving 20 minutes of rest, alternating 1.5-minute periods of fatigue-free and fatigued states, using an electronic hand dynamometer for controlled fatigue induction.
- **Signal Processing:** EMG signals were processed to extract features like mean absolute value (MAV) and root mean square (RMS). The raw EMG data was filtered, mean-corrected, and analyzed using PSD and FFT plots. Accelerometer data was also collected, and its magnitude calculated.



A close up look of the 'Fatigued Trial 4' raw EMG Signal



Trial 4 Fatigue Free - An overlay of Filtered EMG (grey) on the raw EMG signal (blue).

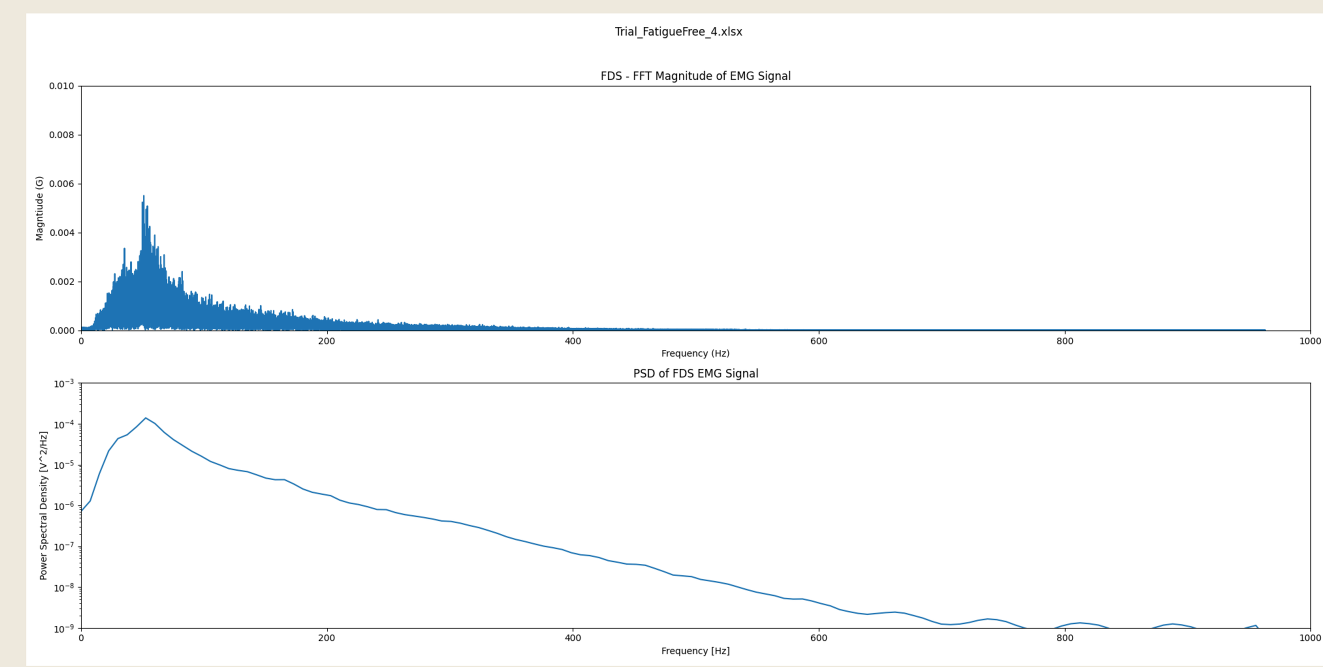


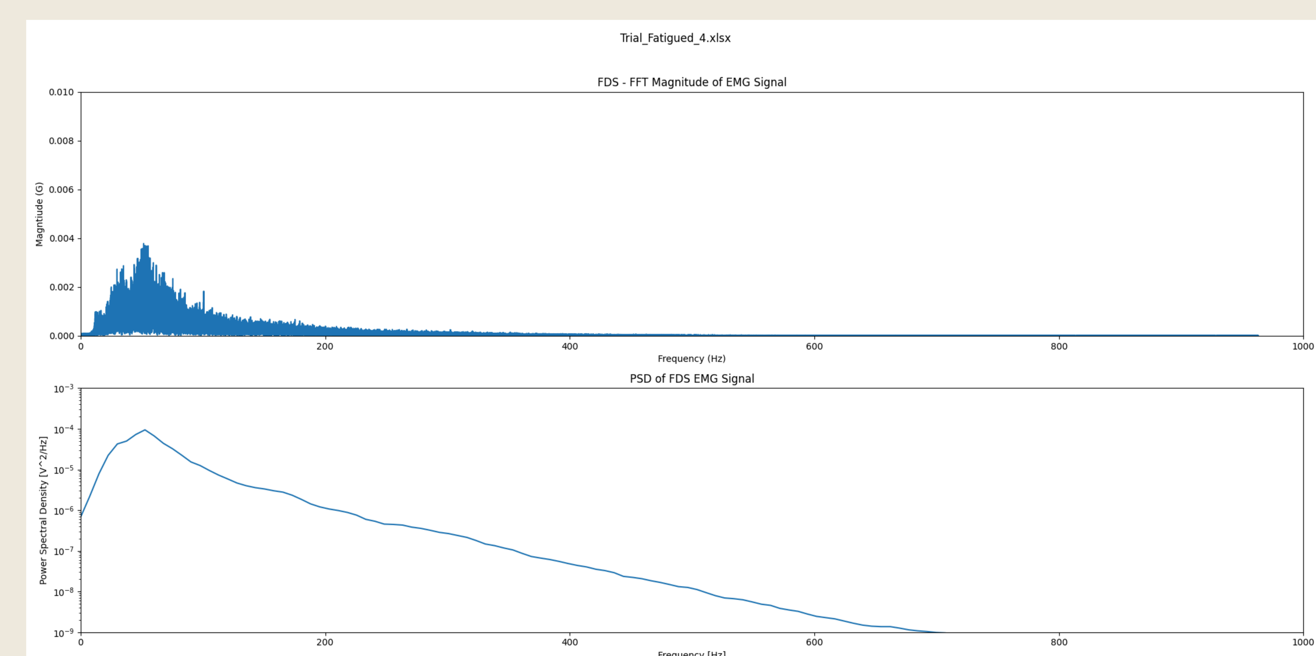
Trial 4 Fatigued - An overlay of Filtered EMG (grey) on the raw EMG signal (blue).

03. Results/Findings

From my study, the following findings emerged:

- Out of 5 trials, only Trials 4 and 5 showed signs of muscle fatigue in the frequency domain analysis.
- No significant muscle fatigue was observed in the time-domain analysis, RMS, or median frequency (MDF), and the accelerometer data remained relatively unchanged.
- Overall, EMG signal amplitudes were consistent across fatigue-free and fatigued trials, indicating limited variation to muscle fatigue.





Fast Fourier Transform (FFT) and Power Spectral Density (PSD) of Trial 4 - Fatigue Free and Fatigued. There is a noticeable decrease in the fatigued trial which is indicative of muscle fatigue. The PSD, however, do not show much difference.

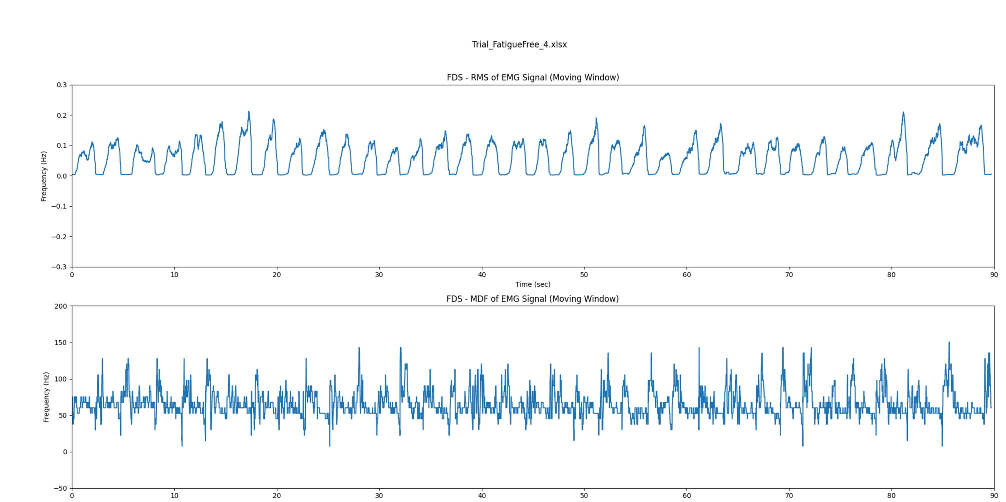
04. Analysis

I analyzed the EMG data using various signal processing techniques to detect muscle fatigue. Here's a summary:

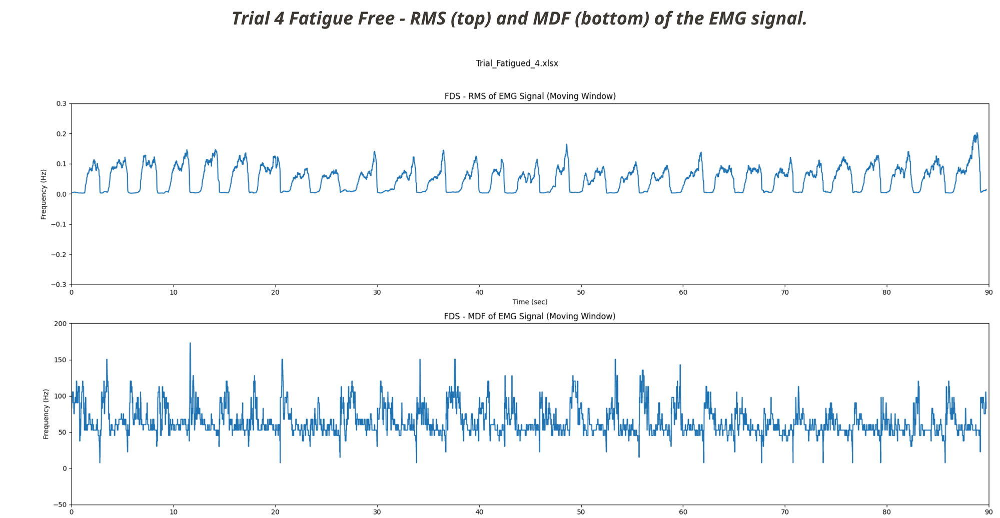
- Data Collection: Collected baseline data using an electronic hand dynamometer. Conducted 5 trials alternating between fatigue-free and fatigued states.
- Signal Processing: Extracted features - MAV, RMS, SD, MDF. Applied filtering, mean-correction, and rectification. Processed accelerometer data and calculated magnitude.
- Key Findings:

  - Frequency Domain: Trial 4 showed a significant drop in frequency indicating muscle fatigue, and trial 5 displayed some signs of fatigue, however, less pronounced.

  - Time Domain: The amplitude remained consistent and RMS and MDF showed no significant fatigue signs.



Trial 4 Fatigue Free - RMS (top) and MDF (bottom) of the EMG signal.



Trial 4 Fatigued - RMS (top) and MDF (bottom) of the EMG signal.

05. Conclusion

In this study, I assessed wearable EMG sensors for monitoring muscle fatigue. Here are the key findings and future directions:

Key Findings:

1. Frequency Domain: Fatigue signs were detected in trials 4 and 5.
2. Time Domain: EMG signal amplitudes remained stable, and RMS and MDF graphs did not show significant fatigue patterns.

Future Directions:

1. Extended Participant Pool: Test with a larger, more diverse group to verify findings and enhance reliability.
2. Enhanced Signal Processing: Explore advanced methods to detect subtle fatigue signs more effectively.
3. Real-time Monitoring: Develop systems using machine learning and deep learning for real-time fatigue prediction to improve worker safety and productivity.

Related literature

1. J. Park, M. Zahabi, D. Kaber, J. Ruiz, and He Helen Huang, "Evaluation of Activities of Daily Living Tesbeds for Assessing Prosthetic Device Usability," 2020 IEEE International Conference on Human-Machine Systems (ICHMS), Sep. 2020, doi: https://doi.org/10.1109/ichms49158.2020.9209553.
2. A. Phinyomark, E. Campbell, and E. Scheme, "Surface Electromyography (EMG) Signal Processing, Classification, and Practical Considerations," Series in BioEngineering, pp. 3–29, Nov. 2019, doi: https://doi.org/10.1007/978-981-13-9097-5_1.
3. H. A. Yousif et al., "Assessment of Muscles Fatigue Based on Surface EMG Signals Using Machine Learning and Statistical Approaches: A Review," IOP Conference Series: Materials Science and Engineering, vol. 705, p. 012010, Dec. 2019, doi: https://doi.org/10.1088/1757-899x/705/1/012010.